

Exercise 08
Soft Constraints and Robust MPC

Exercise 1 **Soft constraints**

Consider the following optimization problem with soft constraints

$$\begin{aligned} \min_{x, \epsilon_i} \quad & f(x) + \sum_{i=1}^r l_i(\epsilon_i) \\ \text{s.t.} \quad & g_i(x) \leq \epsilon_i, \quad i = 1, \dots, r, \\ & \epsilon_i \geq 0, \quad i = 1, \dots, r, \end{aligned} \tag{1}$$

where $x \in \mathbb{R}^n$, $\epsilon_i \in \mathbb{R}$, and $f(x)$, $g_i(x)$, $l_i(x)$, $i = 1, \dots, r$, are scalar-valued convex functions.

- 1) Give a possible form for the penalty functions $l_i(\epsilon_i)$, $i = 1, \dots, r$, such that it is a *smooth exact penalty function*. State the requirements for exactness.
- 2) For $n = 2$, consider the following specific problem:

$$f(x) = \|x\|_2^2, \quad g_1(x) = [-1 \ 0] x + 1, \quad g_2(x) = [0 \ 1] x + 1.$$

Compute the requirements from part 1) for this specific problem and give an example penalty function that satisfies the requirements.

Hint: Determine the primal optimal solution graphically.

Exercise 2 **Robust invariant set**

Consider the uncertain system $x(k+1) = \frac{1}{2}x(k) + w(k)$ under the state constraint $-10 \leq x \leq 10$ and subject to a disturbance bounded to lie in the set $|w| \leq 1$. Compute the maximum robust invariant set